SAIDS — PYQ Question Mapper

Statistics for Artificial Intelligence & Data Science

CSE-AIML Sem V · C Scheme (R-2019) · Dec 2022 – May 2025

PAPER LEGEND

Dec 2022	Dec 2023	May 2024	Dec 2024	May 2025 (AIML)	May 2025 (IT)	IAE QB-1	IAE QB-2

QUESTIONS PER MODULE

Module I	Statistical Foundations, Covariance & Sampling	**43 Qs**
Module II	Statistical Inference, Estimation & Resampling	**45 Qs**
Module III	Regression Models, Regularization & Nonlinear Models	**13 Qs**
Module IV	Classification & Supervised Learning	**7 Qs**
Module V	Unsupervised Learning & Dimensionality Reduction	**8 Qs**
Module VI	Model Evaluation, Bayesian Inference & Ethics	**3 Qs**

Module I Statistical Foundations, Covariance & Sampling

Syllabus Topics: *Data types (categorical, numerical, ordinal) · Descriptive stats (central tendency, dispersion, skewness, kurtosis) · Data viz (histograms, boxplots, scatter plots, pair plots, QQ plots) · Probability distributions (Normal, Binomial, Poisson, Exponential, Uniform, Beta, Gamma, Weibull) · Covariance, correlation, correlation matrices · Sampling (simple random, stratified, cluster) · Central Limit Theorem (CLT) & Law of Large Numbers · Empirical CDF · Data preprocessing (outlier treatment, encoding, normalization)*

Total: 43 questions · Dec 2022: 7 · Dec 2023: 6 · May 2024: 7 · Dec 2024: 8 · May 2025 (AIML): 6 · May 2025 (IT): 5 · IAE QB-1: 4

1. **[Dec 2022 | Q1a]** Find standard deviation of average temperatures over 5-day period: 19, 21, 18, 24, 12 *(Numerical)*

2. **[Dec 2022 | Q1b]** Normal distribution (μ=30, σ=4): find P(x<40) and P(30<x<35) *(Numerical)*

3. **[Dec 2022 | Q1e]** What do you mean by correlation and regression? Explain with example *(Theory)*

4. **[Dec 2022 | Q2a]** Find correlation coefficient from Age (X) vs Glucose Level (Y) data (6 subjects) *(Numerical)*

5. **[Dec 2022 | Q3a]** What is the use of scatter plot and box plot? *(Theory)*

6. **[Dec 2022 | Q6a]** Binomial distribution: insurance agent, P(living 30+ yrs)=2/3 for 5 people — P(all 5), P(≥3), P(exactly 2) *(Numerical)*

7. **[Dec 2022 | Q6b]** Short notes (any two): Central Limit Theorem, Standard Error *(Short notes)*

8. **[Dec 2023 | Q1d]** Short note on Random sampling *(Short note)*

9. **[Dec 2023 | Q1e]** What is the Empirical CDF function? *(Short note)*

10. **[Dec 2023 | Q2a]** Frequency distribution table for 30 orange weights; find class mark, range, number of intervals, lowest-frequency interval *(Numerical)*

11. **[Dec 2023 | Q3b]** Concept of correlation; how is it different from regression? *(Theory)*

12. **[Dec 2023 | Q4a]** Normal distribution: P(car speed > 100 km/hr) with μ=90, σ=10 *(Numerical)*

13. **[Dec 2023 | Q4b]** Explain Numerical and Categorical data types with appropriate examples *(Theory)*

14. **[May 2024 | Q1d]** What do you mean by correlation and regression? Explain with example *(Theory)*

15. **[May 2024 | Q2a]** Normal distribution (μ=30, σ=4): find P(x<40), P(x>21), P(30<x<35) *(Numerical)*

16. **[May 2024 | Q2b]** Poisson distribution: vehicles at 300/hr — P(none in 1 min), expected in 2 min, P(expected number passes in 2 min) *(Numerical)*

17. **[May 2024 | Q3a]** Normal distribution: P(battery charge lasts 50–70 hrs) for μ=50, σ=15 *(Numerical)*

18. **[May 2024 | Q4a]** Explain QQ plots in detail; show how scatterplots explore relationships between variables *(Theory)*

19. **[May 2024 | Q6a]** Explain Binomial distribution in detail; Type I error concept in bottle volume test *(Theory + Numerical)*

20. **[May 2024 | Q6b]** Short notes (any 2): Chi-square distribution, Weibull distribution, Stem & Leaf Plot, Box Plot *(Short notes)*

21. **[Dec 2024 | Q1a]** Explain percentiles and Boxplots with example *(Short note)*

22. **[Dec 2024 | Q1b]** Illustrate Central Limit Theorem with a neat diagram *(Short note)*

23. **[Dec 2024 | Q1d]** What is a Histogram? Give its applications *(Short note)*

24. **[Dec 2024 | Q2b]** Explain Normal and Poisson Distribution; GMAT scores (μ=527, σ=112): P(score>500) and score for top 5% *(Theory + Numerical)*

25. **[Dec 2024 | Q4b]** Mean, SD, standard error and Confidence Interval; compute all for a 10-data-point sample *(Theory + Numerical)*

26. **[Dec 2024 | Q5b]** Continuous Probability Distribution & PDF; Uniform distribution on [0,20]: P(5≤X≤15), P(X<10), P(X>18) *(Numerical)*

27. **[Dec 2024 | Q6b]** Short note: Histogram and Scatterplot with example *(Short note)*

28. **[Dec 2024 | Q6d]** Short note: Weibull Distribution *(Short note)*

29. **[May 2025 (AIML) | Q1c]** Explain the difference between Stratified and Cluster Sampling *(Short note)*

30. **[May 2025 (AIML) | Q1e]** Define standard deviation and interquartile range with examples *(Short note)*

31. **[May 2025 (AIML) | Q2a]** Find the correlation coefficient from given Experience (X) vs Salary (Y) data *(Numerical)*

32. **[May 2025 (AIML) | Q4a]** Z-score: sample mean=250, population mean=260, SD=40; calculate z-score for sample mean *(Numerical)*

33. **[May 2025 (AIML) | Q4b]** Frequency distribution table for 50 college students' weekly study hours *(Numerical)*

34. **[May 2025 (AIML) | Q6b]** Short notes (any 2): Chi-square distribution, Weibull distribution, Stem & Leaf Plot, Box Plot *(Short notes)*

35. **[May 2025 (IT) | Q1b]** Explain Poisson probability distribution with one example *(Short note)*

36. **[May 2025 (IT) | Q1d]** Discuss Quantitative Data, Descriptive Statistics, Variables, Categorical data *(Theory)*

37. **[May 2025 (IT) | Q1e]** Why is sampling necessary? Describe population and sample mean *(Short note)*

38. **[May 2025 (IT) | Q2b]** Binomial distribution: insurance agent, P(living 30+ yrs)=2/3 for 5 people — P(all 5), P(≥3), P(exactly 2) *(Numerical)*

39. **[May 2025 (IT) | Q6b]** Short notes: Scatter diagram, Sample space, Discrete probability distribution *(Short notes)*

40. **[IAE QB-1 | A2]** What is mean? Find the mean of the data: 4, 6, 8, 10 *(Numerical)*

41. **[IAE QB-1 | B1]** Find the mean, median and mode for the following marks: 10, 20, 20, 30, 40 *(Numerical)*

42. **[IAE QB-1 | B5]** A university has 2000 students. Explain an appropriate sampling technique to select a sample; use CLT to explain how statistical inference can be drawn from the sample *(Theory)*

43. **[IAE QB-1 | B6]** Binomial distribution: A factory produces bulbs with P(defective)=0.1. If n=5 bulbs are selected, find P(exactly 2 are defective) *(Numerical)*

Module II Statistical Inference, Estimation & Resampling

Syllabus Topics: *Point estimates · Bias, Variance, MSE · MLE & Method of Moments · Confidence intervals (mean, proportion, variance) · Hypothesis testing: z-test, t-test, Chi-square, F-test, ANOVA (1-way, 2-way) · Nonparametric tests: Mann-Whitney U, Kruskal-Wallis, Friedman, Wilcoxon, Sign test · Bootstrap CI & resampling · Likelihood Ratio Tests (LRT) · Type I & II errors · p-value*

Total: 45 questions · Dec 2022: 8 · Dec 2023: 7 · May 2024: 7 · Dec 2024: 8 · May 2025 (AIML): 7 · May 2025 (IT): 5 · IAE QB-1: 3

1. [Dec 2022 | Q1c] Discuss Bootstrapping vs. re-sampling *(Short note)*

Answer:

- Bootstrap: A resampling technique that draws samples WITH replacement from the original observed data to estimate the sampling distribution of a statistic (PDF Q.13)
- Bootstrap CI steps (PDF Q.13): (1) Collect original sample; (2) Draw multiple bootstrap samples with replacement; (3) Compute statistic (mean/median) for each; (4) Arrange values; (5) Take 2.5th and 97.5th percentiles for 95% CI
- Example from PDF (Q.13): Waiting times {8,10,12,14,16}; 5 bootstrap samples give means: 11.6, 13.2, 11.6, 12.4, 12.8; 95% Bootstrap CI ≈ [11.6, 13.2]
- Example from PDF (Q.13): Marks {40,45,50,55,60,70}; 8 bootstrap samples computed; 95% Bootstrap CI ≈ [47.5, 57.5]
- Permutation Test: A resampling method for hypothesis testing; randomly shuffles group labels WITHOUT replacement to generate the null distribution and compute p-value (PDF Q.13)
- Permutation steps (PDF Q.13): (1) Compute D_obs (observed group difference); (2) Shuffle group labels; (3) Recompute statistic; (4) p-value = (permutations with $|D| \geq D_{obs}$) / total permutations
- Example from PDF (Q.13): Group A[10,12,14], B[16,18,20]; D_obs=6; 180 out of 10000 permutations had $|D| \geq 6$; p=180/10000=0.018 < 0.05 → Reject H_0
- Key difference (PDF Q.14): Bootstrap → resamples WITH replacement, estimates uncertainty, output is Confidence Interval; Permutation → shuffles WITHOUT replacement, performs hypothesis testing, output is p-value

2. [Dec 2022 | Q1d] School principal: state null and alternative hypotheses (teachers claim students use computer 3.2 hrs/day) *(Theory)*

Answer:

- H_0 (Null Hypothesis): $\mu = 3.2$ hrs/day — The default claim; teachers' assertion is true; students use computers exactly 3.2 hours/day on average (PDF Q.6)
- H_1 (Alternative Hypothesis): $\mu \neq 3.2$ hrs/day — The research claim; the actual average daily usage is different from 3.2 hours (two-tailed test)
- Two-tailed test: Tests deviation in EITHER direction (more OR less than 3.2 hrs); use when no prior direction is assumed (PDF Q.6)
- Right-tailed alternative (if testing for MORE usage): H_1: $\mu > 3.2$ hrs/day
- Left-tailed alternative (if testing for LESS usage): H_1: $\mu < 3.2$ hrs/day
- Significance level: Choose $\alpha = 0.05$; reject H_0 if test statistic exceeds ±z-critical or ±t-critical at that level
- Decision rule (PDF Q.6): If |test statistic| > critical value OR p-value < α → Reject H_0; conclude actual usage differs significantly from 3.2 hrs/day
- We never "accept" H_0; we either REJECT H_0 (evidence is strong) or FAIL TO REJECT H_0 (insufficient evidence)

3. [Dec 2022 | Q2b] Explain why ANOVA is used; solve One-way ANOVA for Observations A, B, C (5×3 table) *(Theory + Numerical)*

Answer:

- ANOVA (Analysis of Variance): Used to compare means of 3 or more groups simultaneously; avoids inflating Type I error that would result from running multiple t-tests (PDF Q.11)
- Key formula: F = MSB/MSW = [SSB/(k−1)] / [SSW/(N−k)] where k = number of groups, N = total observations (PDF Q.11)
- 8-step procedure from PDF (Q.11): (1) State H_0: $\mu_1=\mu_2=\mu_3$; H_1: at least one mean differs; (2) Compute group means and Grand Mean; (3) $SSB = \Sigma n_i(\bar{x}_i - \bar{x}grand)^2$
- Steps continued (PDF Q.11): (4) $SSW = \Sigma(x_{ij} - \bar{x}_i)^2$; (5) SST = SSB + SSW; (6) MSB = SSB/(k−1); MSW = SSW/(N−k); (7) F = MSB/MSW; (8) Compare F with F-critical at α=0.05
- Between-group variance (MSB): Measures variation DUE TO the treatment/factor; Within-group variance (MSW): Measures random error/noise inside each group
- Example from PDF (3 fertilizers, Q.11): k=3 groups, N=15 total; F=27 > F-critical=4.26 → REJECT H_0; fertilizers produce significantly different plant growth
- Example from PDF (teaching methods, Q.11): 3 teaching methods; F=75 → REJECT H_0; different methods produce significantly different exam scores
- Decision rule: If F > F-critical → Reject H_0 (at least one group mean differs significantly); if F ≤ F-critical → Fail to reject H_0

4. [Dec 2022 | Q3a] Explain Type I & Type II error in detail *(Theory)*

Answer:

- Type I Error (False Positive): Rejecting H_0 when H_0 is actually TRUE; this is an incorrect rejection (PDF Q.7)
- Probability of Type I Error = α (the significance level, typically set at 0.05 by the researcher before testing)
- Example of Type I Error (PDF Q.7): Declaring a new drug effective when it actually has no therapeutic effect on patients
- Type II Error (False Negative): Failing to reject H_0 when H_0 is actually FALSE; this is an incorrect non-rejection (PDF Q.7)
- Probability of Type II Error = β
- Example of Type II Error (PDF Q.7): Declaring a drug ineffective when it actually works — missing a true treatment effect in clinical trials
- Power of the test = 1 − β = the probability of correctly rejecting a false H_0 (detecting a real effect)
- Trade-off (PDF Q.7): Decreasing α (stricter significance) increases β, and vice versa; to reduce BOTH errors simultaneously, increase the sample size n

5. [Dec 2022 | Q3b] Kruskal-Wallis test: 4 manufacturing facilities, exam scores after machining training, at 95% confidence — are scores same? *(Numerical)*

Answer:

- Kruskal-Wallis Test: Non-parametric alternative to one-way ANOVA; used when data is NOT normally distributed or is ordinal (PDF Q.12)
- H_0: All k groups (facilities) come from the same distribution — exam scores are the same across all 4 facilities
- H_1: At least one group has a different distribution — scores differ significantly between at least two facilities
- Step 1: Combine ALL observations from all 4 groups and rank them from 1 to N (assign average ranks for ties)
- Step 2: Compute rank sum R_i for each facility group separately; verify $R_1+R_2+R_3+R_4 = N(N+1)/2$
- Step 3: $H = [12/N(N+1)] \times \Sigma(R_i^2/n_i) - 3(N+1)$ where N = total observations, n_i = observations in group i (PDF Q.12)
- Step 4: df = k−1 = 4−1 = 3; at α=0.05 (95% confidence), find χ^2-critical from chi-square table with df=3; χ^2-critical(3, 0.05) = 7.815
- Decision: If H > 7.815 → REJECT H_0 (scores differ significantly across facilities); reference from PDF (3 teaching methods, N=12): H=9.846 > 5.991 → Reject H_0

6. [Dec 2022 | Q4a] t-score: sample mean=290, expected mean=300, SD=50, n=15 — find t-score *(Numerical)*

Answer:

- t-test is used when population standard deviation σ is UNKNOWN and sample size may be small; uses t-distribution (PDF Q.9)
- Formula (PDF Q.9): $t = (\bar{x} - \mu) / (s/\sqrt{n})$ where $\bar{x}$ = sample mean, μ = hypothesized population mean, s = sample SD, n = sample size
- Degrees of freedom: df = n − 1
- Given values: $\bar{x} = 290$, $\mu = 300$, $s = 50$, $n = 15$
- Standard error: $s/\sqrt{n} = 50/\sqrt{15} = 50/3.873 = 12.91$
- $t = (290 - 300) / 12.91 = -10 / 12.91 = -0.775$
- t-score = −0.775 with df = n − 1 = 14
- Reference from PDF (Q.9): Task time example — μ=50, $\bar{x}$=54, s=5, n=10 → t=(54−50)/(5/√10)=4/1.581=2.53, df=9; since 2.53 > t-critical=2.262, reject H_0

7. [Dec 2022 | Q5a] z-test: pea plant pods (μ=145, σ=100, $\bar{x}$=147) — state hypotheses, test statistic at α=0.05 *(Numerical)*

Answer:

- z-test is used when population standard deviation σ is KNOWN and/or sample size n ≥ 30 (PDF Q.8)
- Formula (PDF Q.8): $Z = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$
- H_0: μ = 145 (mean number of pods per plant = 145)
- H_1: μ ≠ 145 (two-tailed test — testing if mean pods differ from 145 in either direction)
- Given: $\bar{x}$=147, μ_0=145, σ=100; Standard error = $\sigma/\sqrt{n}$; $Z = (147-145)/(100/\sqrt{n}) = 2/(100/\sqrt{n})$
- At α=0.05 (two-tailed), z-critical = ±1.96; if |Z| > 1.96 → Reject H_0
- Reference from PDF (battery example, Q.8): n=100, $\bar{x}$=11.8, μ=12, σ=0.5 → Z=(11.8−12)/(0.5/√100)=−0.2/0.05=−4; since |−4|>1.96, REJECT H_0
- Conclusion: If |Z| > 1.96, sufficient evidence to reject claim; if |Z| ≤ 1.96, fail to reject H_0 (pods do not differ significantly from 145)

8. [Dec 2022 | Q6b] Short note: Confidence Interval *(Short note)*

Answer:

- Confidence Interval (CI): A range of values derived from sample data estimated to contain the true population parameter with a stated confidence level (PDF Q.5)
- CI for population mean (σ known): $CI = \bar{x} \pm z^*(\sigma/\sqrt{n})$ using z-distribution
- CI for population mean (σ unknown): $CI = \bar{x} \pm t^*(s/\sqrt{n})$ using t-distribution at df = n−1
- CI for proportion: $CI = \hat{p} \pm z^*\sqrt{[\hat{p}(1-\hat{p})/n]}$
- CI for variance: $CI = [(n-1)s^2/\chi^2_upper, (n-1)s^2/\chi^2_lower]$
- Solved example from PDF (Q.5): data {10,12,14,16,18}; $\bar{x}$=14, s=3.16, n=5; SE=3.16/√5=1.414; t^*=2.776 (df=4, α/2=0.025)
- 95% CI = 14 ± 2.776×1.414 = 14 ± 3.927 = (10.073, 17.927)

• Interpretation: We are 95% confident the true population mean lies between 10.073 and 17.927; narrower CI = more precision; CI width decreases as n increases

9. [Dec 2023 | Q1a] Short note on hypothesis testing *(Short note)*

Answer:

• Hypothesis Testing: A statistical procedure to determine if sample evidence is sufficient to reject a null claim about a population parameter (PDF Q.6)
• Step 1: State H_0 (null hypothesis — default claim, contains "=") and H_1 (alternative — what we want to prove)
• Step 2: Choose significance level α (typically 0.05); Step 3: Select appropriate test statistic — z-test (σ known), t-test (σ unknown), χ^2-test (categorical), F-test/ANOVA (multiple groups)
• Step 4: Compute test statistic from sample data; Step 5: Find critical value from table OR compute p-value
• Step 6: Decision — if |test statistic| > critical value OR p-value < α → REJECT H_0; otherwise FAIL TO REJECT H_0
• Step 7: Draw conclusion in context of the original problem
• Types of tests (PDF Q.6): Two-tailed (H_1: $\mu \neq \mu_0$), Right-tailed (H_1: $\mu > \mu_0$), Left-tailed (H_1: $\mu < \mu_0$)
• We never "accept" H_0; we either REJECT it (strong evidence against it) or FAIL TO REJECT it (insufficient evidence to overturn it)

10. [Dec 2023 | Q1b] What is Fisher's exact test? *(Short note)*

Answer:

• Note: Fisher's exact test is not explicitly covered in the provided Chapter 2 PDF notes; the following is based on syllabus context
• Fisher's exact test: Used to analyze the association between two categorical variables in a 2×2 contingency table when sample sizes are SMALL
• Unlike Chi-square (which uses approximations), Fisher's test computes the EXACT probability of observing the given cell frequencies (or more extreme) under H_0 of no association
• When to use: When at least one expected cell frequency < 5 — this violates the Chi-square test assumption and makes Chi-square unreliable
• Uses the hypergeometric distribution to calculate the exact probability of the observed table configuration
• H_0: The two categorical variables are independent (no association); H_1: They are associated
• Advantage: No minimum sample size requirement; statistically accurate even for very small samples
• Commonly used in: Medical research, clinical trials, genetics studies, and biological experiments with small group sizes

11. [Dec 2023 | Q2b] Purpose of one-way ANOVA; key components: dependent variable, independent variable, factors *(Theory)*

Answer:

• One-way ANOVA: Tests whether the means of 3+ groups defined by ONE factor are significantly different; avoids Type I error inflation from multiple t-tests (PDF Q.11)
• Key component 1 — Dependent variable: The outcome being measured (e.g., exam scores, crop yield, student performance)
• Key component 2 — Independent variable (Factor): The grouping variable with k levels (e.g., teaching method A/B/C, fertilizer type 1/2/3)
• Key component 3 — Between-group variance: MSB = SSB/(k−1); SSB = $\Sigma n_i(\bar{X}_i - \bar{X}grand)^2$; measures variation DUE TO the factor/treatment
• Key component 4 — Within-group variance: MSW = SSW/(N−k); SSW = $\Sigma(x_{ij} - \bar{X}_i)^2$; measures random error/noise WITHIN each group
• F-statistic = MSB/MSW; a large F means group differences are large relative to within-group variation
• H_0: $\mu_1=\mu_2=\ldots=\mu_k$ (all group means equal); H_1: at least one group mean differs; compare F with F-critical at α=0.05
• Example from PDF (fertilizers, Q.11): F=27 > F-critical=4.26 → REJECT H_0; significant difference in plant growth across fertilizer types

12. [Dec 2023 | Q3a] Standard error of estimate for household children data; 95% confidence interval *(Numerical)*

Answer:

• Standard Error (SE) = $s/\sqrt{n}$; measures how far the sample mean is expected to vary from the true population mean (PDF Q.5)
• A smaller SE means a more precise estimate; SE decreases as sample size n increases (SE ∝ $1/\sqrt{n}$)
• 95% CI formula: $\bar{X} \pm t^*(s/\sqrt{n})$ where t^* = t-critical value at df=n−1 and α/2=0.025
• Solved example from PDF (Q.5): data {10,12,14,16,18}; n=5, $\bar{X}$=14, s=3.16
• SE = $s/\sqrt{n}$ = $3.16/\sqrt{5}$ = 3.16/2.236 = 1.414
• t-critical (df=4, two-tailed, α=0.05) = 2.776
• 95% CI = 14 ± 2.776 × 1.414 = 14 ± 3.927 = (10.073, 17.927)
• Interpretation (PDF Q.5): We are 95% confident the true mean number of children per household lies between 10.073 and 17.927

13. [Dec 2023 | Q5a] t-test: Duracell CEO claims μ=300h; sample n=20, $\bar{X}$=270h, s=50h — reject claim? *(Numerical)*

Answer:

• t-test used when population SD σ is UNKNOWN; formula: $t = (\bar{X} - \mu)/(s/\sqrt{n})$ (PDF Q.9)

- H_0: μ = 300h (CEO's claim is true — batteries last 300 hours on average)
- H_1: μ < 300h (one-tailed, LEFT-tailed — testing if batteries last LESS than claimed)
- Given: n=20, $\bar{x}$=270h, s=50h; df = n−1 = 19
- s/√n = 50/√20 = 50/4.472 = 11.18
- t = (270 − 300) / 11.18 = −30 / 11.18 = −2.68
- At α=0.05 (one-tailed, left), t-critical = −1.729 (df=19); since |−2.68| > 1.729 → REJECT H_0
- Conclusion: Sufficient evidence to reject CEO's claim; batteries do NOT last 300 hours on average (reference PDF Q.9: same t-test procedure)

14. [Dec 2023 | Q6a] z-test: pea plant pods (μ=145, σ=100, sample $\bar{x}$=147) — state hypotheses and test statistic *(Numerical)*

Answer:

- z-test is used when population SD σ is KNOWN; formula: $Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$ (PDF Q.8)
- H_0: μ = 145 (mean number of pods per pea plant = 145)
- H_1: μ ≠ 145 (two-tailed — testing if mean pods differ from 145 in EITHER direction)
- Given: $\bar{x}$=147, μ_0=145, σ=100; Standard error = σ/√n
- Z = (147 − 145) / (100/√n) = 2 / (100/√n); value depends on given n
- At α=0.05 (two-tailed), z-critical = ±1.96; if |Z| > 1.96 → REJECT H_0
- Reference from PDF (battery example, Q.8): n=100, $\bar{x}$=11.8, μ=12, σ=0.5 → Z=(11.8−12)/(0.5/10)=−4; since |−4|>1.96, REJECT H_0 (batteries do not last claimed 12 hours)
- If |Z| ≤ 1.96 → FAIL TO REJECT H_0; insufficient evidence that mean pods differ from 145

15. [Dec 2023 | Q6b] What is Chi-Square Test and in what situations is it commonly used? *(Theory)*

Answer:

- Chi-square (χ^2) test: A non-parametric test for categorical data; formula: $\chi^2 = \Sigma[(O-E)^2/E]$ where O=observed, E=expected frequency (PDF Q.10)
- Expected frequency: E = (Row Total × Column Total) / Grand Total for each cell in the contingency table
- Use case 1 — Test of Independence: Tests if two categorical variables are associated (e.g., gender vs. brand preference)
- Use case 2 — Goodness of Fit: Tests if observed data follows a hypothesized distribution
- df = (rows−1)(cols−1) for test of independence; compare χ^2 with χ^2-critical from chi-square table
- Example from PDF (brand preference, Q.10): χ^2=2.00; χ^2-critical(df=2, α=0.05)=5.991; since 2.00<5.991, FAIL to reject H_0 (no significant brand preference difference)
- Example from PDF (gender vs online classes, Q.10): χ^2=4 > χ^2-critical=3.841 → REJECT H_0; gender and online class preference are significantly associated
- Decision: If χ^2 > χ^2-critical → Reject H_0 (variables are dependent/associated); if χ^2 ≤ χ^2-critical → Fail to reject H_0

16. [May 2024 | Q1a] Define Confidence Interval *(Short note)*

Answer:

- Confidence Interval (CI): A range of values derived from sample data estimated to contain the true population parameter with a stated confidence level (e.g., 95%) (PDF Q.5)
- CI = point estimate ± margin of error
- CI for mean (σ known): CI = $\bar{x} \pm z^*(\sigma/\sqrt{n})$ using z-distribution (e.g., z^*=1.96 for 95%)
- CI for mean (σ unknown): CI = $\bar{x} \pm t^*(s/\sqrt{n})$ using t-distribution at df = n−1
- CI for proportion: CI = $\hat{p} \pm z^*\sqrt{[\hat{p}(1-\hat{p})/n]}$; CI for variance: $[(n-1)s^2/\chi^2_upper, (n-1)s^2/\chi^2_lower]$
- Solved example from PDF (Q.5): data {10,12,14,16,18}; $\bar{x}$=14, s=3.16, n=5, t^*=2.776 → 95% CI = (10.073, 17.927)
- Interpretation: A 95% CI means if we repeated sampling 100 times, 95 of those intervals would contain the true parameter
- Narrower CI = more precision (larger n); wider CI = more uncertainty (smaller n)

17. [May 2024 | Q1b] Standard Error of sample mean: μ=$12, σ=$3, n=30 — estimate and interpret SE *(Numerical)*

Answer:

- Standard Error (SE) of the sample mean measures how much the sample mean varies from the true population mean (PDF Q.5)
- Formula: SE = σ/√n (when population σ is known) or SE = s/√n (when using sample SD)
- Given: μ=$12, σ=$3, n=30
- SE = σ/√n = 3/√30 = 3/5.477 = 0.547
- Interpretation: The sample mean is expected to vary by approximately $0.55 from the true population mean across repeated samples of size 30
- SE decreases as sample size increases: SE ∝ 1/√n; doubling n reduces SE by a factor of √2 ≈ 1.414
- SE is used to construct Confidence Intervals: CI = $\bar{x} \pm z^*(SE)$ or $\bar{x} \pm t^*(SE)$
- A small SE indicates the sample mean is a reliable and precise estimate of the population mean μ

18. [May 2024 | Q1c] What is hypothesis testing? Explain Type I and Type II errors *(Short note)*

Answer:

- Hypothesis testing: A formal procedure to evaluate claims about population parameters using sample data (PDF Q.6)
- Steps (PDF Q.6): (1) State H_0 (null) and H_1 (alternative); (2) Choose α; (3) Compute test statistic; (4) Compare with critical value or p-value; (5) Decision and conclusion
- Type I Error (α, False Positive): Rejecting H_0 when H_0 is actually TRUE (PDF Q.7)
- Probability of Type I Error = α (the significance level chosen by the researcher, typically 0.05)
- Example (PDF Q.7): Declaring a new drug effective when it actually has no therapeutic effect
- Type II Error (β, False Negative): Failing to reject H_0 when H_0 is actually FALSE (PDF Q.7)
- Probability of Type II Error = β; Power of the test = 1 − β = probability of correctly detecting a real effect
- Trade-off (PDF Q.7): Decreasing α increases β and vice versa; to reduce BOTH simultaneously, increase sample size n

19. [May 2024 | Q1e] What is analysis of variance (ANOVA)? Explain its usage *(Short note)*

Answer:

- ANOVA (Analysis of Variance): A statistical test to compare means of THREE OR MORE groups simultaneously (PDF Q.11)
- Why ANOVA: Running multiple t-tests (e.g., AB, AC, BC for 3 groups) inflates Type I error; ANOVA uses a single F-test, keeping α controlled
- Key formula: F = MSB/MSW where MSB = Between-group variance = SSB/(k−1); MSW = Within-group variance = SSW/(N−k)
- H_0: $\mu_1=\mu_2=\ldots=\mu_k$ (all group means equal); H_1: at least one mean differs; if F > F-critical → Reject H_0
- One-way ANOVA: ONE factor/independent variable; e.g., testing 3 teaching methods on exam scores
- Two-way ANOVA: TWO factors; can detect main effects and INTERACTION effects between factors
- Example from PDF (fertilizers, Q.11): k=3, N=15; F=27 > F-critical=4.26 → REJECT H_0; significant difference in plant growth across fertilizer types
- Example from PDF (teaching methods, Q.11): F=75 → REJECT H_0; teaching methods produce significantly different exam scores

20. [May 2024 | Q3b] z-test: score change (μ=80, σ=10, n=38, $\bar{x}$=88) at α=0.05 — evidence for claim? *(Numerical)*

Answer:

- z-test is used when population SD σ is KNOWN; formula: $Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$ (PDF Q.8)
- H_0: μ = 80 (mean score remains at 80 after training)
- H_1: μ > 80 (one-tailed, RIGHT-tailed — testing if training IMPROVED scores)
- Given: $\bar{x}$=88, μ_0=80, σ=10, n=38
- $\sigma/\sqrt{n} = 10/\sqrt{38} = 10/6.164 = 1.622$
- Z = (88 − 80) / 1.622 = 8/1.622 = 4.93
- At α=0.05 (one-tailed, right), z-critical = 1.645; since 4.93 > 1.645 → REJECT H_0
- Conclusion: Strong statistical evidence at α=0.05 that the mean score has significantly increased above 80 (reference: same z-test procedure, PDF Q.8)

21. [May 2024 | Q4b] One-way ANOVA numerical: four samples A, B, C, D — identify differences *(Numerical)*

Answer:

- One-way ANOVA with 4 groups (k=4): Tests if means of groups A, B, C, D differ significantly (PDF Q.11)
- Step 1: H_0: μA=μB=μC=μD (all means equal); H_1: at least one group mean differs
- Step 2: Compute each group mean $\bar{x}$A, $\bar{x}$B, $\bar{x}$C, $\bar{x}$D and the Grand Mean $\bar{x}$grand = (ΣAll values)/N
- Step 3: $SSB = \Sigma n_i(\bar{x}_i - \bar{x}grand)^2$; measures variation BETWEEN groups; df_between = k−1 = 3
- Step 4: $SSW = \Sigma(x_{ij} - \bar{x}_i)^2$; measures variation WITHIN groups; df_within = N−k = N−4
- Step 5: MSB = SSB/3; MSW = SSW/(N−4); F = MSB/MSW
- Step 6: Find F-critical from F-distribution table at α=0.05 with df=(3, N−4)
- Decision: If F > F-critical → REJECT H_0; at least one group mean is significantly different (same procedure as PDF Q.11 examples)

22. [May 2024 | Q5a] F-Test: F=2.38, df=(8,3) — find F-table value; reject null at α=0.05 (one-tailed)? *(Numerical)*

Answer:

- F-test as used in ANOVA (PDF Q.11): F = MSB/MSW; tests if between-group variation is significantly larger than within-group variation
- In ANOVA context: H_0 — all group means are equal; H_1 — at least one mean differs
- F-distribution is right-skewed; critical value depends on numerator df (df_1) and denominator df (df_2)
- Given: F-calculated = 2.38, df_1 = 8, df_2 = 3
- From F-distribution table at α=0.05 (one-tailed) with df=(8,3): F-critical ≈ 8.85
- Since F-calculated = 2.38 < F-critical = 8.85 → FAIL TO REJECT H_0
- Conclusion: No significant difference detected at α=0.05; the group means do not differ significantly

- Note: The standalone F-test for comparing two population variances ($F = s_1^2/s_2^2$) is not explicitly detailed in the provided Chapter 2 PDF notes; the F-statistic in PDF Q.11 is used exclusively in ANOVA

23. [Dec 2024 | Q1c] Explain Null and alternative Hypothesis with example *(Short note)*

Answer:

- H_0 (Null Hypothesis): The default claim; assumes no effect or no difference; always contains "=" (e.g., H_0: $\mu = 50$) (PDF Q.6)
- H_0 is the hypothesis we attempt to disprove with sample evidence
- H_1 (Alternative Hypothesis): The research claim we want to prove using data (PDF Q.6)
- Two-tailed H_1: $\mu \neq 50$ (testing for deviation in EITHER direction)
- Right-tailed H_1: $\mu > 50$ (testing if mean is GREATER than 50)
- Left-tailed H_1: $\mu < 50$ (testing if mean is LESS than 50)
- Example: H_0 — A new drug has no effect ($\mu = \mu_control$); H_1 — The drug has a measurable effect ($\mu \neq \mu_control$)
- Rule (PDF Q.6): We never "accept" H_0; we either REJECT H_0 (when evidence is strong) or FAIL TO REJECT H_0 (when evidence is insufficient)

24. [Dec 2024 | Q1f] What is the F-Test used for? *(Short note)*

Answer:

- F-test: Uses the F-statistic to test hypotheses about variance; F = MSB/MSW in ANOVA context (PDF Q.11)
- Primary use in ANOVA (PDF Q.11): F = MSB/MSW tests if between-group variance is significantly larger than within-group (error) variance
- H_0 in ANOVA: All group means are equal; H_1: At least one mean differs; if F > F-critical → Reject H_0
- MSB = SSB/(k−1) (between-group mean square); MSW = SSW/(N−k) (within-group mean square)
- F-distribution: Right-skewed; depends on numerator df (df_1=k−1) and denominator df (df_2=N−k)
- F-critical from F-distribution table at α=0.05 with the respective df values
- Example from PDF (fertilizers, Q.11): F=27 > F-critical=4.26 → REJECT H_0; fertilizers produce significantly different plant growth
- Example from PDF (teaching methods, Q.11): F=75 → REJECT H_0; teaching methods produce significantly different exam scores

25. [Dec 2024 | Q2a] Chi-Square Test: gender vs. coffee/tea preference contingency table (Male/Female × Coffee/Tea) at α=0.05 *(Numerical)*

Answer:

- Chi-square test of independence: Tests if gender and beverage preference are associated (PDF Q.10)
- H_0: Gender and beverage preference are INDEPENDENT (not associated)
- H_1: Gender and beverage preference are DEPENDENT (associated)
- Step 1: Construct 2×2 contingency table with observed frequencies O (Male/Female × Coffee/Tea)
- Step 2: Compute expected frequency for each cell: E = (Row Total × Column Total) / Grand Total
- Step 3: Compute $\chi^2 = \Sigma[(O-E)^2/E]$ summing over all 4 cells
- Step 4: df = (rows−1)(cols−1) = (2−1)(2−1) = 1; at α=0.05, χ^2-critical (df=1) = 3.841
- Decision: If $\chi^2 > 3.841$ → REJECT H_0 (gender and beverage preference are significantly associated); if $\chi^2 \leq 3.841$ → FAIL TO REJECT H_0 (reference: PDF Q.10 gender vs online classes example χ^2=4 > 3.841 → reject H_0)

26. [Dec 2024 | Q3b] t-Distribution: CEO claims bulb lasts 300 days; n=15, $\bar{x}$=290, s=50 — $P(\bar{x} \leq 290)$ *(Numerical)*

Answer:

- t-test used when population SD σ is UNKNOWN; formula: $t = (\bar{x} - \mu)/(s/\sqrt{n})$ (PDF Q.9)
- H_0: $\mu = 300$ days (CEO's claim is true — bulbs last 300 days on average)
- H_1: $\mu < 300$ days (one-tailed, LEFT-tailed — testing if bulbs last LESS than claimed)
- Given: n=15, $\bar{x}$=290 days, s=50 days; df = n−1 = 14
- Standard error: $s/\sqrt{n} = 50/\sqrt{15} = 50/3.873 = 12.91$
- $t = (290 - 300) / 12.91 = -10/12.91 = -0.775$
- At α=0.05 (one-tailed, left), t-critical = −1.761 (df=14); since $|-0.775| < 1.761$ → FAIL TO REJECT H_0
- Conclusion: Insufficient evidence to reject CEO's claim; observed mean of 290 days is not significantly below 300; $P(\bar{x} \leq 290)$ corresponds to t=−0.775 with df=14, giving p-value > 0.05

27. [Dec 2024 | Q4a] ANOVA: why use it; one-way vs two-way difference; solve one-way ANOVA for 3 groups *(Theory + Numerical)*

Answer:

- Why ANOVA: Comparing 3+ group means using multiple t-tests would inflate Type I error (false positive); ANOVA uses a single F-test to keep α controlled (PDF Q.11)
- One-way ANOVA: Tests effect of ONE factor (independent variable) on dependent variable; k group means are compared

• Two-way ANOVA: Tests effects of TWO factors simultaneously; can also detect INTERACTION effects between factors (does effect of Factor A depend on Factor B?)
• One-way ANOVA steps from PDF (Q.11): (1) H_0/H_1; (2) Compute group means and Grand Mean; (3) $SSB=\Sigma n_i(\bar{X}_i-\bar{X}grand)^2$; (4) $SSW=\Sigma(x_{ij}-\bar{X}_i)^2$; (5) SST=SSB+SSW; (6) MSB=SSB/(k−1); MSW=SSW/(N−k); (7) F=MSB/MSW; (8) Compare with F-critical
• Example from PDF (fertilizers, Q.11): k=3, N=15, F=27 > F-critical=4.26 → REJECT H_0; different fertilizers produce significantly different plant growth
• Example from PDF (teaching methods, Q.11): k=3; F=75 → REJECT H_0; different teaching methods produce significantly different exam scores
• F > F-critical → Reject H_0 (at least one group mean differs significantly); F ≤ F-critical → Fail to reject H_0
• Note: Kruskal-Wallis example from PDF (Q.12): 3 fertilizers, N=9; H=7.2 > 5.991 → significant difference (non-parametric ANOVA alternative)

28. [Dec 2024 | Q5a] Nonparametric vs parametric methods; Friedman Test: THM water cleanup data (12 counties) *(Theory + Numerical)*

Answer:

• Parametric tests: Assume data follows normal distribution; used with interval/ratio scale data; examples from PDF: z-test (Q.8), t-test (Q.9), ANOVA (Q.11)
• Non-parametric tests: Do NOT assume normality; used for ordinal data, non-normal distributions, or small samples; examples from PDF (Q.12): Mann-Whitney U, Kruskal-Wallis
• Mann-Whitney U (PDF Q.12): Non-parametric alternative to independent t-test; compares two groups; ranks all data together; H_0: groups have same distribution
• Example from PDF (Mann-Whitney, Q.12): Teaching A[12,15,14,10,13] vs B[18,20,16,17,19]; U=0 → REJECT H_0; group B scores significantly higher
• Kruskal-Wallis (PDF Q.12): Non-parametric alternative to one-way ANOVA; $H=[12/N(N+1)]\times\Sigma(R_i^2/n_i)-3(N+1)$; compare with χ^2-critical (df=k−1)
• Example from PDF (Kruskal-Wallis, Q.12): 3 fertilizers, N=9; H=7.2 > χ^2-critical=5.991 → REJECT H_0; significant difference in plant growth
• Friedman Test: Non-parametric alternative to two-way/repeated-measures ANOVA; ranks values within each block and compares treatment rank sums
• Note: Specific Friedman test formula and step-by-step procedure are not detailed in the provided Chapter 2 PDF notes; Mann-Whitney U and Kruskal-Wallis are covered in detail (PDF Q.12)

29. [Dec 2024 | Q6a] Short note: Type I and Type II errors in detail *(Short note)*

Answer:

• Type I Error (False Positive): Rejecting H_0 when H_0 is actually TRUE (PDF Q.7)
• Probability of Type I Error = α (the significance level set by the researcher before testing, typically 0.05)
• Example from PDF (Q.7): Declaring a new drug effective when it actually has no therapeutic effect on patients
• Type II Error (False Negative): Failing to reject H_0 when H_0 is actually FALSE (PDF Q.7)
• Probability of Type II Error = β
• Example from PDF (Q.7): Declaring a drug ineffective when it actually works — missing a true treatment effect (false negative)
• Power of the test = 1 − β = probability of correctly detecting a real effect when it truly exists
• Key relationship (PDF Q.7): α and β are inversely related — decreasing α (stricter test) increases β; to minimize BOTH simultaneously, increase the sample size n

30. [Dec 2024 | Q6c] Short note: Bootstrapping vs Re-sampling *(Short note)*

Answer:

• Bootstrap CI: Resampling technique that draws samples WITH replacement from original observed data (PDF Q.13)
• Bootstrap steps (PDF Q.13): (1) Collect original sample; (2) Draw multiple samples with replacement; (3) Compute statistic for each; (4) Find 2.5th and 97.5th percentile for 95% CI
• Bootstrap example from PDF (Q.13): Marks {40,45,50,55,60,70}; 8 bootstrap samples computed; 95% Bootstrap CI ≈ [47.5, 57.5]
• Another example from PDF (Q.13): Waiting times {8,10,12,14,16}; 5 bootstrap means: 11.6, 13.2, 11.6, 12.4, 12.8; 95% CI ≈ [11.6, 13.2]
• Permutation Test: Resampling for hypothesis testing; shuffles labels WITHOUT replacement to generate null distribution (PDF Q.13)
• Permutation example from PDF (Q.13): A[10,12,14], B[16,18,20]; D_obs=6; 180/10000 permutations had $|D|\geq 6$; p=0.018 < 0.05 → REJECT H_0
• Key comparison from PDF (Q.14): Bootstrap → WITH replacement, estimates uncertainty, output=CI; Permutation → WITHOUT replacement, tests hypothesis, output=p-value
• Bootstrap answers: "What range of values is plausible for the parameter?"; Permutation answers: "Is the observed difference due to chance?"

31. [May 2025 (AIML) | Q1a] What is hypothesis testing? Explain Type I and Type II errors *(Short note)*

Answer:

- Hypothesis Testing: A formal procedure to evaluate claims about population parameters using sample data (PDF Q.6)
- Steps (PDF Q.6): (1) State H_0 (null — default, "=") and H_1 (alternative — what we want to prove); (2) Choose α; (3) Collect data; (4) Compute test statistic
- Steps continued: (5) Compare test statistic with critical value OR compute p-value; (6) Make decision: Reject H_0 if evidence is sufficient; (7) State conclusion in context
- Type I Error (False Positive): Rejecting H_0 when H_0 is actually TRUE (PDF Q.7); P(Type I) = α
- Example of Type I Error: Concluding a drug treatment works when it actually has no real effect on patients
- Type II Error (False Negative): Failing to reject H_0 when H_0 is actually FALSE (PDF Q.7); P(Type II) = β
- Example of Type II Error: Declaring a drug ineffective when it actually has a real therapeutic effect
- Power = 1 − β; increasing sample size n reduces both α and β simultaneously (PDF Q.7)

32. [May 2025 (AIML) | Q1b] What is Fisher's exact test? *(Short note)*

Answer:

- Note: Fisher's exact test is not explicitly covered in the provided Chapter 2 PDF notes; the following is based on syllabus context
- Fisher's exact test: Used for 2×2 contingency tables when sample sizes are SMALL (expected cell frequencies < 5)
- Unlike Chi-square (which uses approximations), Fisher's test computes the EXACT probability of observing the given table or more extreme, under H_0 of no association
- Uses the hypergeometric distribution to compute exact probabilities for the table configuration
- When to use: When at least one expected cell frequency < 5 — this violates the Chi-square minimum frequency assumption
- H_0: The two categorical variables are independent; H_1: They are associated
- Advantage over Chi-square: No approximation; statistically accurate for very small samples — no minimum sample size requirement
- Applications: Medical research, clinical trials, genetics studies, and biological experiments where small group sizes are unavoidable

33. [May 2025 (AIML) | Q2b] Chi-Square Test: gender vs. online/in-store shopping (n=200) at α=0.05 *(Numerical)*

Answer:

- Chi-square test of independence applied to gender vs. shopping preference (PDF Q.10)
- H_0: Gender and shopping mode (online/in-store) are INDEPENDENT — no association between them
- H_1: Gender and shopping mode are DEPENDENT — they are significantly associated
- Step 1: Construct 2×2 contingency table (Male/Female × Online/In-store) with observed frequencies O; total n=200
- Step 2: Compute expected frequency for each cell: E = (Row Total × Column Total) / 200
- Step 3: $\chi^2 = \Sigma[(O-E)^2/E]$ computed across all 4 cells; formula from PDF Q.10
- Step 4: df = (2−1)(2−1) = 1; at α=0.05, χ^2-critical (df=1) = 3.841
- Decision: If $\chi^2 > 3.841 \rightarrow$ REJECT H_0; gender and shopping mode are significantly associated (reference: PDF Q.10 gender vs online classes: $\chi^2=4 > 3.841 \rightarrow$ reject H_0)

34. [May 2025 (AIML) | Q3a] Explain the concept of p-value in hypothesis testing *(Theory)*

Answer:

- p-value: The probability of observing a test statistic as extreme as (or more extreme than) the one calculated, ASSUMING H_0 is true (PDF Q.6 context)
- Decision rule: If p-value < α → REJECT H_0 (result is statistically significant); if p-value ≥ α → FAIL TO REJECT H_0
- Example from PDF (permutation test, Q.13): Group A[10,12,14], B[16,18,20]; D_obs=6; 180 out of 10000 permutations had $|D| \geq 6$
- p-value = 180/10000 = 0.018; since 0.018 < 0.05 → REJECT H_0; the observed group difference is unlikely to be due to chance
- Interpretation: A p-value of 0.018 means there is only a 1.8% probability of observing this result if H_0 were truly correct
- Common thresholds: p < 0.05 (significant), p < 0.01 (highly significant), p < 0.001 (very highly significant)
- A small p-value provides STRONG evidence against H_0; a large p-value means weak evidence against H_0
- Important: p-value does NOT measure the size or practical importance of the effect; it does NOT equal the probability that H_0 is true

35. [May 2025 (AIML) | Q3b] Kruskal-Wallis test: aptitude scores of Grade A, B, C students at 95% confidence level *(Numerical)*

Answer:

- Kruskal-Wallis test: Non-parametric alternative to one-way ANOVA for 3+ independent groups (PDF Q.12)
- H_0: All 3 grade groups (A, B, C) come from the same distribution — no significant difference in aptitude scores
- H_1: At least one grade group has a different distribution of aptitude scores
- Step 1: Combine all observations (N total) and assign ranks 1 to N; assign average ranks for tied values
- Step 2: Compute rank sum R_i for each grade group; verify: $R_1+R_2+R_3 = N(N+1)/2$
- Step 3: $H = [12/N(N+1)] \times \Sigma(R_i^2/n_i) - 3(N+1)$ (PDF Q.12)
- Step 4: df = k−1 = 3−1 = 2; at α=0.05 (95% confidence), χ^2-critical (df=2) = 5.991
- Example from PDF (3 teaching methods, N=12, Q.12): R1=10, R2=26, R3=42; H=9.846 > 5.991 → REJECT H_0; aptitude scores differ significantly across Grade A, B, C

36. [May 2025 (AIML) | Q5a] z-test: pharmaceutical blood pressure drug (μ=10mmHg, σ=5, n=30, $\bar{x}$=12) at α=0.05 — state H_0/H_1, test stat, conclusion *(Numerical)*

Answer:

- z-test is used when population SD σ is KNOWN; formula: $Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$ (PDF Q.8)
- H_0: μ = 10 mmHg (drug reduces blood pressure by exactly 10 mmHg)
- H_1: μ > 10 mmHg (one-tailed, RIGHT-tailed — testing if the drug reduces BP by MORE than 10 mmHg)
- Given: n=30, $\bar{x}$=12, σ=5, μ_0=10
- $\sigma/\sqrt{n} = 5/\sqrt{30} = 5/5.477 = 0.913$
- $Z = (12 - 10) / 0.913 = 2/0.913 = 2.19$
- At α=0.05 (one-tailed, right), z-critical = 1.645; since 2.19 > 1.645 → REJECT H_0
- Conclusion: Sufficient evidence at α=0.05 that the drug reduces blood pressure by MORE than 10 mmHg (reference: same z-test procedure from PDF Q.8: battery example Z=−4 → reject H_0)

37. [May 2025 (AIML) | Q6a] Two-way ANOVA: concept; difference from one-way ANOVA; assumptions; Friedman's test as nonparametric alternative *(Theory)*

Answer:

- One-way ANOVA (PDF Q.11): Tests effect of ONE factor on the dependent variable; F = MSB/MSW; k group means compared
- Two-way ANOVA: Tests effects of TWO factors simultaneously; can also detect INTERACTION effects between Factor A and Factor B
- Key difference: One-way uses ONE independent variable; two-way uses TWO independent variables and reveals whether their interaction is significant
- Interaction effect in two-way ANOVA: Does the effect of Factor A depend on the level of Factor B? (e.g., does teaching method effectiveness depend on student grade level?)
- Assumptions: Normality of residuals, homogeneity of variances across groups, independence of observations
- One-way ANOVA 8-step procedure from PDF (Q.11): H_0/H_1 → means → SSB → SSW → SST → MSB/MSW → F → compare F with F-critical
- Example from PDF (fertilizers, Q.11): F=27 > F-critical=4.26 → REJECT H_0; significant effect of fertilizer type
- Friedman Test: Non-parametric alternative to two-way repeated-measures ANOVA; ranks values within each block; detailed procedure not covered in the provided Chapter 2 PDF notes

38. [May 2025 (IT) | Q1c] What is statistical inference? *(Short note)*

Answer:

- Statistical inference: The process of drawing conclusions about a POPULATION based on analysis of a SAMPLE drawn from it (PDF context: Q.1–Q.5)
- Two main branches: (1) Estimation and (2) Hypothesis Testing
- Point Estimation (PDF Q.1): Provides a single value estimate of population parameter; $\hat{\theta}$ estimates θ; uses MLE (Q.3) and Method of Moments (Q.4)
- Properties of good estimators (PDF Q.1): Unbiasedness ($E(\hat{\theta}) = \theta$), Consistency (converges to θ as n→∞), Efficiency (minimum variance), Sufficiency
- Interval Estimation (PDF Q.5): Provides a RANGE with stated confidence; 95% CI = $\bar{x} \pm t^*(s/\sqrt{n})$; solved example: {10,12,14,16,18} → CI = (10.073, 17.927)
- Hypothesis Testing (PDF Q.6–Q.12): Tests claims about parameters using z-test, t-test, χ^2-test, ANOVA, non-parametric tests
- Key principle: Sample statistics ($\bar{x}$, s, $\hat{p}$) estimate population parameters (μ, σ, p) with quantified uncertainty
- MSE = Variance + $Bias^2$ (PDF Q.2); a good estimator minimizes both bias and variance for accurate inference

39. [May 2025 (IT) | Q2a] Null and alternate hypothesis; explain Type I and Type II error *(Theory)*

Answer:

- H_0 (Null Hypothesis): The default claim; assumes no effect or equality; always contains "=" (e.g., H_0: μ = μ_0) (PDF Q.6)
- H_1 (Alternative Hypothesis): The research claim we want to prove; can be two-tailed (≠), right-tailed (>), or left-tailed (<) (PDF Q.6)
- We never "accept" H_0; we either REJECT H_0 (strong evidence against it) or FAIL TO REJECT H_0 (insufficient evidence)
- Type I Error (False Positive): Rejecting H_0 when it is actually TRUE (PDF Q.7); Probability = α
- Example of Type I Error: Concluding a treatment is effective when it actually has no real effect
- Type II Error (False Negative): Failing to reject H_0 when it is actually FALSE (PDF Q.7); Probability = β
- Example of Type II Error: Failing to detect a real treatment effect (declaring an effective drug ineffective)
- Power = 1 − β; decision table: Correct decision → reject false H_0 (Power) or retain true H_0; Incorrect → Type I error or Type II error

40. [May 2025 (IT) | Q5a] t-test: pizza shop cheese weights (n=7) — state hypotheses, test statistic; reject at 10%, 5%, 1% level? *(Numerical)*

Answer:

- t-test formula: $t = (\bar{x} - \mu)/(s/\sqrt{n})$; degrees of freedom = n − 1 = 7 − 1 = 6 (PDF Q.9)
- Step 1: State H_0 (e.g., μ = target cheese weight) and H_1 (e.g., $\mu \neq$ target weight, two-tailed)
- Step 2: Calculate sample mean $\bar{x}$ and sample standard deviation s from the 7 cheese weights provided
- Step 3: Compute t-statistic = $(\bar{x} - \mu) / (s/\sqrt{7})$
- Step 4: At 10% significance level (df=6): t-critical = 1.943; REJECT H_0 if $|t| > 1.943$
- Step 5: At 5% significance level (df=6): t-critical = 2.447; REJECT H_0 if $|t| > 2.447$
- Step 6: At 1% significance level (df=6): t-critical = 3.707; REJECT H_0 if $|t| > 3.707$
- Conclusion: Reject H_0 at those significance levels where |t-calculated| exceeds the corresponding t-critical (reference: same procedure PDF Q.9: t=2.53 at df=9 > t-critical=2.262 → reject H_0)

41. [May 2025 (IT) | Q6a] What is Sign Test? Explain Wilcoxon Signed-Rank Test *(Theory)*

Answer:

- Note: The specific Sign test procedure and Wilcoxon Signed-Rank test details are not explicitly covered in the provided Chapter 2 PDF notes; the following is based on syllabus context
- Sign Test: Simplest non-parametric test for single sample or paired data; tests if median differs from a hypothesized value
- Sign Test procedure: Compute difference $d_i = x_i - \mu_0$ for each observation; count the number of + and − signs (ignore zeros)
- Test statistic B = number of positive signs; compare B with binomial critical value for (n, p=0.5)
- If B is too large or too small (two-tailed), REJECT H_0 — the median differs from the hypothesized value
- Wilcoxon Signed-Rank Test: More powerful than Sign test; uses both SIGN and MAGNITUDE of differences
- Wilcoxon steps: (1) Compute $d_i = x_i - \mu_0$; (2) Rank $|d_i|$ ignoring zeros; (3) Compute W^+ (sum of positive-sign ranks) and W^- (sum of negative-sign ranks); (4) T = min(W^+, W^-); (5) Compare T with Wilcoxon critical value from table
- Used for: Paired samples or single sample vs hypothesized median when data is non-normal (non-parametric alternative to one-sample t-test)

42. [May 2025 (IT) | Q6b] Short note: Kruskal-Wallis Test *(Short note)*

Answer:

- Kruskal-Wallis Test: Non-parametric alternative to one-way ANOVA; used when data is NOT normally distributed or is ordinal (PDF Q.12)
- When to use: (1) Data distribution is non-normal; (2) Data is ordinal/ranked; (3) Sample sizes are small
- H_0: All k groups come from the same distribution; H_1: At least one group distribution differs
- Formula (PDF Q.12): $H = [12/N(N+1)] \times \Sigma(R_i^2/n_i) - 3(N+1)$ where R_i = rank sum of group i, n_i = group size, N = total observations
- df = k − 1; compare H with χ^2-critical from chi-square distribution table at α=0.05
- Example from PDF (3 fertilizers, N=9, Q.12): R1=6, R2=15, R3=24; H=7.2 > χ^2-critical(df=2)=5.991 → REJECT H_0; significant difference in plant growth
- Example from PDF (3 teaching methods, N=12, Q.12): R1=10, R2=26, R3=42; H=9.846 > 5.991 → REJECT H_0; scores differ significantly
- Advantage: No normality assumption required; can be applied to ordinal/ranked data; suitable for small samples

43. [IAE QB-1 | A7] What is bias of an estimator? Explain its significance *(Theory)*

Answer:

- Bias of an estimator $\hat{\theta}$: Bias($\hat{\theta}$) = E($\hat{\theta}$) − θ where E($\hat{\theta}$) is the expected value and θ is the true unknown parameter (PDF Q.1)
- Unbiased estimator: Bias = 0, meaning E($\hat{\theta}$) = θ; example: sample mean $\bar{x}$ is unbiased for μ since E($\bar{x}$) = μ (PDF Q.1)
- Biased estimator: Bias > 0 means $\hat{\theta}$ systematically OVERESTIMATES θ; Bias < 0 means it UNDERESTIMATES θ
- MSE (Mean Squared Error) = Variance($\hat{\theta}$) + [Bias($\hat{\theta}$)]^2 (PDF Q.2)
- An unbiased estimator with high variance may have a HIGHER MSE than a slightly biased estimator with very low variance
- Properties of a good estimator (PDF Q.1): Unbiasedness, Consistency ($\hat{\theta} \to \theta$ as $n \to \infty$), Efficiency (minimum variance), Sufficiency
- Significance: Low bias ensures the estimator does not systematically miss the true parameter value in repeated sampling
- Ideal estimator: Unbiased (E($\hat{\theta}$)=θ) AND minimum variance → Best Linear Unbiased Estimator (BLUE)

44. [IAE QB-1 | A8] What is Maximum Likelihood Estimation (MLE)? *(Theory)*

Answer:

- MLE: Finds the parameter value $\hat{\theta}$ that maximizes the likelihood function L(θ) = P(data|θ), making the observed data most probable (PDF Q.3)
- Steps from PDF (Q.3): (1) Write likelihood function L(θ); (2) Take natural log-likelihood: ln L(θ); (3) Differentiate w.r.t. θ; (4) Set derivative = 0; (5) Solve for $\hat{\theta}$_MLE
- Example from PDF (Bernoulli, Q.3): Coin tosses H,H,T,H,T → $L(p)=p^3(1-p)^2$; ln L=3ln(p)+2ln(1−p); d/dp=0 → $\hat{p}$_MLE = 3/5 = 0.6
- Example from PDF (Binomial, Q.3): n=10 items, x=2 defectives → $L(p)=C(10,2)p^2(1-p)^8$; $\hat{p}$_MLE = x/n = 2/10 = 0.2
- Advantage 1 (PDF Q.3): Consistency — as $n \to \infty$, $\hat{\theta}$_MLE converges to the true θ
- Advantage 2 (PDF Q.3): Asymptotic efficiency — achieves minimum variance in large samples

- Advantage 3 (PDF Q.3): Invariance property — if $\hat{\theta}_{MLE}$ is MLE of θ, then $g(\hat{\theta}_{MLE})$ is MLE of $g(\theta)$
- Limitations (PDF Q.3): May be biased in small samples; computationally intensive for complex models; requires correct specification of the likelihood function

45. [IAE QB-1 | B3] z-test: A manufacturer claims batteries last μ=50h. Test at 5% significance: n=36, $\bar{x}$=53h, σ=9h — state H_0/H_1, compute test statistic, and give conclusion *(Numerical)*

Answer:

- z-test is used when population SD σ is KNOWN; formula: $Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$ (PDF Q.8)
- H_0: $\mu = 50$h (manufacturer's claim is true — batteries last exactly 50 hours)
- H_1: $\mu \neq 50$h (two-tailed test — testing if batteries last a DIFFERENT amount than claimed)
- Given: n=36, $\bar{x}$=53h, σ=9h, μ_0=50h
- $\sigma/\sqrt{n} = 9/\sqrt{36} = 9/6 = 1.5$
- $Z = (53 - 50) / 1.5 = 3/1.5 = 2.0$
- At α=0.05 (two-tailed), z-critical = ±1.96; since $|2.0| > 1.96 \rightarrow$ REJECT H_0
- Conclusion: Sufficient statistical evidence at 5% significance to reject the manufacturer's claim; batteries do NOT last exactly 50 hours (reference: PDF Q.8 battery example: n=100, $\bar{x}$=11.8, μ=12, σ=0.5 → Z=−4 → same approach)

Module III Regression Models, Regularization & Nonlinear Models

Syllabus Topics: *Simple & multiple linear regression · LLSR derivation · Model assumptions · Residual analysis · Multicollinearity (VIF) · Polynomial & interaction effects · Nonlinear regression · Regularization: Ridge, Lasso, Elastic Net · Spline regression & LOESS · GLM: Logistic regression (binary), Poisson (count data) · Model evaluation: MAE, MSE, RMSE, R^2, Adjusted R^2*

Total: 13 questions · Dec 2022: 2 · Dec 2023: 2 · May 2024: 1 · Dec 2024: 1 · May 2025 (AIML): 2 · May 2025 (IT): 4 · IAE QB-1: 1

1. **[Dec 2022 | Q4b]** Multiple regression: find relation between GPA of students and number of study hours and height *(Numerical)*

2. **[Dec 2022 | Q5b]** Simple linear regression equation + coefficient of determination for Hour–Temperature data *(Numerical)*

3. **[Dec 2023 | Q1c]** Short note on Simple Linear Regression *(Short note)*

4. **[Dec 2023 | Q5b]** Explain Linear Least Square Regression (LLSR) with advantages and disadvantages *(Theory)*

5. **[May 2024 | Q5b]** Find simple linear regression equation and coefficient of determination (R^2) for given X–Y data *(Numerical)*

6. **[Dec 2024 | Q3a]** Explain single and Multiple linear regression with example and suitable plot *(Theory)*

7. **[May 2025 (AIML) | Q1d]** Explain Linear Regression and its Applications *(Short note)*

8. **[May 2025 (AIML) | Q5b]** Find the simple linear regression equation for Time–Growth data *(Numerical)*

9. **[May 2025 (IT) | Q3b]** Differentiate between Simple Linear Regression and Multiple Regression *(Theory)*

10. **[May 2025 (IT) | Q4a]** What is Logistic Regression? *(Theory)*

11. **[May 2025 (IT) | Q4b]** Find simple linear regression equation for Bill–Tip data *(Numerical)*

12. **[May 2025 (IT) | Q5b]** Derive Simple Linear Regression parameters to predict the value of the output *(Derivation)*

13. **[IAE QB-1 | A10]** What is multicollinearity in regression? Explain its effect on the model *(Theory)*

Module IV Classification & Supervised Learning

Syllabus Topics: *Classification metrics (accuracy, precision, recall, F1, ROC-AUC, PR curves) · Logistic regression, odds ratio · Naive Bayes · k-NN classifier · Decision Trees & pruning · Ensemble methods: Bagging, Random Forest, Gradient Boosting · SVM (linear & kernel) · Handling imbalanced data: SMOTE · Confusion matrix analysis*

Total: 7 questions · IAE QB-2: 7

1. [IAE QB-2 | A1] What is the difference between precision and recall? Mention one situation where recall is preferred over precision *(Theory)*

2. [IAE QB-2 | A2] A classification model produces TP=40, FP=10, FN=5, TN=45. Calculate the precision of the model *(Numerical)*

3. [IAE QB-2 | A3] What is SMOTE? Why is it used when handling an imbalanced classification dataset? *(Theory)*

4. [IAE QB-2 | A4] What is the kernel trick in SVM? State its purpose *(Theory)*

5. [IAE QB-2 | B1] Classification model for fraudulent transactions (imbalanced dataset): a) Why can accuracy be misleading? b) How do precision, recall, and F1-score provide better information? c) Which metric to prioritize to detect as many fraudulent transactions as possible? *(Theory)*

6. [IAE QB-2 | B2] A Decision Tree gives 95% training accuracy but 78% testing accuracy. a) Identify the likely problem. b) Explain how pruning can help. c) State one advantage of Random Forest over a single Decision Tree *(Theory)*

7. [IAE QB-2 | B3] Explain the working principle of SVM: a) What is a hyperplane? b) What is the margin and why is maximizing it important? c) Explain the kernel trick with one suitable example *(Theory)*

Note: No questions from PYQs map to this module. All 7 questions above are from the IAE Question Bank (QB-2) only. Classification algorithms remain high-priority for upcoming exams.

Module V Unsupervised Learning & Dimensionality Reduction

Syllabus Topics: *PCA (covariance matrix, Eigen decomposition, explained variance) · LDA (supervised dim. reduction) · Kernel PCA & ICA (conceptual) · Clustering: K-means, Hierarchical, DBSCAN · Cluster evaluation: silhouette score, Davies-Bouldin index · Dimensionality reduction: t-SNE, UMAP · Outlier detection: Z-score, IQR, Isolation Forest, LOF*

Total: 8 questions · Dec 2024: 1 · IAE QB-2: 7

1. [Dec 2024 | Q1e] How to detect outliers? *(Short note)*

2. [IAE QB-2 | A5] What is the purpose of explained variance in PCA? How does it help in selecting the number of principal components? *(Theory)*

3. [IAE QB-2 | A6] Mean of feature=50, SD=5. Calculate Z-score for observation value=65. Is it an outlier using |Z|>3 criterion? *(Numerical)*

4. [IAE QB-2 | A7] What is the main difference between PCA and LDA in terms of whether they use class labels? *(Theory)*

5. [IAE QB-2 | A8] What is the silhouette score in clustering? What does a high silhouette score indicate about cluster quality? *(Theory)*

6. [IAE QB-2 | B4] PCA on correlated features: a) Explain how PCA uses covariance matrix and eigen decomposition to reduce dimensionality. b) What is explained variance? c) Why is retaining the first few principal components useful for ML models? *(Theory)*

7. [IAE QB-2 | B5] Customer transaction dataset: a) Explain K-means clustering for identifying customer groups. b) State one limitation of K-means and how DBSCAN addresses it. c) Explain how silhouette score evaluates cluster quality *(Theory)*

8. [IAE QB-2 | B6] High-dimensional data with potential outliers: a) Explain purpose of t-SNE or UMAP for visualizing high-dimensional data. b) Explain IQR method for outlier detection. c) Name one tree-based method for outlier detection and briefly explain its idea *(Theory)*

Note: Only 1 question from PYQs (outlier detection). 7 additional questions are from the IAE Question Bank (QB-2). PCA, K-means, DBSCAN, t-SNE, and silhouette score are high-potential exam topics.

Module VI Model Evaluation, Bayesian Inference & Ethics

Syllabus Topics: *Model selection: AIC, BIC · Bias-variance trade-off · Cross-validation: k-fold, LOOCV · Calibration curves & Brier score · Bayesian fundamentals: prior, likelihood, posterior · MAP vs MLE · Conjugate priors · Naive Bayes from Bayesian view · Interpretability: SHAP, LIME, PDP, ICE · Fairness in ML · Ethical considerations · Data privacy · Reproducible ML (MLflow, W&B)*

Total: 3 questions · May 2025 (IT): 1 · IAE QB-2: 2

1. [May 2025 (IT) | Q1a] What do you mean by random variables? Explain Bayes theorem *(Theory)*

2. [IAE QB-2 | A9] Differentiate between AIC and BIC as model selection criteria. Mention one key difference in how they penalize model complexity *(Theory)*

3. [IAE QB-2 | A10] A model has high training accuracy but significantly lower validation accuracy. Identify the likely issue using the bias–variance trade-off *(Theory)*

Note: Only 1 question from PYQs (Bayes theorem). 2 additional questions are from the IAE Question Bank (QB-2). AIC/BIC and bias-variance trade-off are now covered via IAE QB. Cross-validation, SHAP/LIME, and ML fairness remain untested.

KEY TAKEAWAYS FOR EXAM PREP

1. **Modules I & II dominate every paper (~65–70% of marks)**
   Every paper has hypothesis testing questions (z-test, t-test, ANOVA, Chi-square), normal/Poisson distribution numericals, confidence intervals, and descriptive statistics. Prioritise these two modules above all others.
2. **Module III (Regression) appears in every paper**
   Expect at least 1–2 numerical (find the SLR equation for given data) and 1 theory question (explain LLSR / multiple regression). Know how to compute regression coefficients, R^2, and residuals.
3. **Hypothesis testing numericals are repeated every exam**
   z-test, t-test, one-way ANOVA, and Chi-square independence tests have numerical problems in every paper. Memorise the formulas: test statistic, degrees of freedom, and decision rule.
4. **Nonparametric tests are increasing (Dec 2024 → May 2025)**
   Friedman test, Kruskal-Wallis test, and Wilcoxon Signed-Rank test appeared in the newer papers. Prepare these as short-note or theory questions, not just numericals.
5. **Modules IV, V, VI: IAE QB covers what PYQs miss**
   Classification (SVM, Decision Trees, Random Forest, SMOTE), PCA/clustering (K-means, DBSCAN, silhouette score), and model evaluation (AIC, BIC, bias-variance) are covered in the IAE Question Bank but absent from PYQs. Prepare SVM working principle, PCA with eigen decomposition, and confusion-matrix numericals as priority topics.
6. **Recurring short-note topics across papers**
   Histogram, Box Plot, Stem & Leaf, Weibull distribution, Type I/II errors, Fisher's exact test, and Bootstrapping appear as short notes in multiple papers. Prepare 5-line answers for each.